

## Math 211 Second Exam (Version A: Sections 1, 53, 54)

7 July 2018

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*Question 1: 10 points, questions 2-7: 15 points each.*

1. Let  $f(x)$  be a positive continuous function on  $[a, b]$  and let  $D$  be the region under the graph  $y = f(x)$  and above the interval  $[a, b]$ . Show that  $\iint_D x \, dA = \int_a^b x f(x) \, dx$ .
2. Evaluate  $\int_0^2 \int_{2y}^4 y(1+x^3)^{-\frac{1}{2}} \, dx \, dy$ .
3. Let  $E$  be the solid that lies inside the sphere  $x^2 + y^2 + z^2 = 2z$  and outside the sphere  $x^2 + y^2 + z^2 = 3$ . Give a description of  $E$  in spherical coordinates.
4. Let  $E$  be the solid below the cone  $z = \sqrt{x^2 + y^2}$  and above the ring  $\frac{1}{4} \leq x^2 + y^2 \leq 1$  in the  $xy$ -plane. Find the mass of  $E$  if the density function is  $\rho(x, y, z) = 5y^2$ .
5. Evaluate  $\int_C x e^y \, ds$  where  $C$  is the line segment from  $P(1, 0)$  to  $Q(\frac{1}{4}, 1)$ .
6. Let  $R$  be the region in the first quadrant that is bounded by the ellipse  $9x^2 + 4y^2 = 1$ . Evaluate  $\iint_R y \, dA$  by making an appropriate change of variables.
7. Find the area of the part of the surface  $z = 16 + 2x - y^2$  that lies above the triangle with vertices  $(0, 0)$ ,  $(-3, 1)$ , and  $(1, 1)$ .

# SOLUTIONS

1. The region  $D$  can be described as a Type I region:  $D = \{(x, y) \mid a \leq x \leq b, 0 \leq y \leq f(x)\}$ , which gives

$$\iint_D x \, dA = \int_a^b \int_0^{f(x)} x \, dy \, dx = \int_a^b x \left[ y \right]_0^{f(x)} dx = \int_a^b x f(x) \, dx.$$

2. Note that  $\int_0^2 \int_{2y}^4 y(1+x^3)^{-\frac{1}{2}} \, dx \, dy = \iint_D y(1+x^3)^{-\frac{1}{2}} \, dA$  where

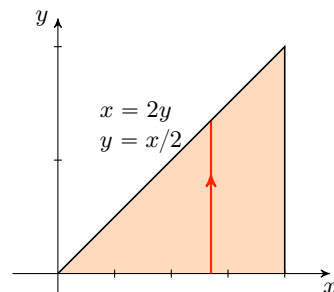
$$D = \{(x, y) \mid 0 \leq y \leq 2, 2y \leq x \leq 4\}.$$

Using a sketch of  $D$  we may describe it as a Type II region:

$$D = \{(x, y) \mid 0 \leq x \leq 4, 0 \leq y \leq x/2\}$$

which gives

$$\iint_D y(1+x^3)^{-\frac{1}{2}} \, dA = \int_0^4 \int_0^{x/2} y(1+x^3)^{-\frac{1}{2}} \, dy \, dx = \int_0^4 \frac{1}{8} x^2 (1+x^3)^{-\frac{1}{2}} \, dx = \frac{1}{12} (1+x^3)^{\frac{1}{2}} \Big|_0^4 = \frac{1}{12} (\sqrt{65} - 1).$$



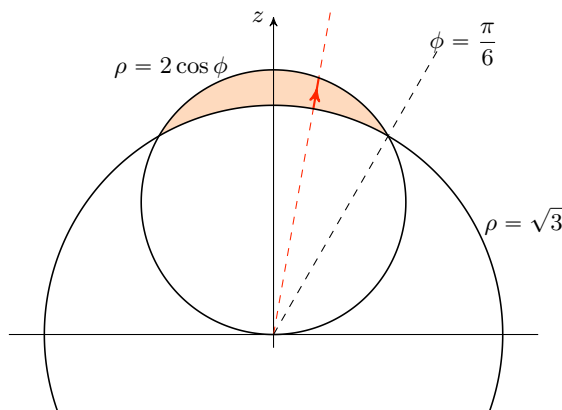
3. A sketch of a vertical cross section of the solid  $E$  is shown.

The two given spheres are represented by the spherical equations  $\rho = 2 \cos \phi$  ( $0 \leq \phi \leq \pi/2$ ) and  $\rho = \sqrt{3}$  respectively, thus their intersection is at

$$2 \cos \phi = \sqrt{3} \Leftrightarrow \cos \phi = \frac{\sqrt{3}}{2} \Rightarrow \phi = \pi/6$$

Now a description of  $E$  in spherical coordinates is

$$E = \{(\rho, \theta, \phi) \mid 0 \leq \theta \leq 2\pi, 0 \leq \phi \leq \frac{\pi}{6}, \sqrt{3} \leq \rho \leq 2 \cos \phi\}.$$



4. In cylindrical coordinates,  $E = \{(r, \theta, z) \mid 0 \leq \theta \leq 2\pi, \frac{1}{2} \leq r \leq 1, 0 \leq z \leq r\}$ . Thus

$$\begin{aligned} m &= \iiint_E 5y^2 \, dV = \int_0^{2\pi} \int_{\frac{1}{2}}^1 \int_0^r 5r^3 \sin^2 \theta \, dz \, dr \, d\theta = \int_0^{2\pi} \int_{\frac{1}{2}}^1 5r^4 \sin^2 \theta \, dr \, d\theta \\ &= \int_0^{2\pi} \left[ r^5 \right]_{\frac{1}{2}}^1 \sin^2 \theta \, d\theta = \frac{31}{32} \int_0^{2\pi} \frac{1}{2} (1 - \cos 2\theta) \, d\theta = \frac{31}{32} \left[ \frac{1}{2} \theta - \frac{1}{4} \sin 2\theta \right]_0^{2\pi} = \frac{31\pi}{32}. \end{aligned}$$

5. Using the vector  $\overrightarrow{PQ} = \langle -\frac{3}{4}, 1 \rangle$ , a parametric representation of the line segment  $C$  is

$$x(t) = 1 - \frac{3}{4}t, \quad y(t) = t, \quad 0 \leq t \leq 1,$$

using which we obtain

$$\int_C x e^y \, ds = \int_0^1 \left(1 - \frac{3}{4}t\right) e^t \sqrt{\left(-\frac{3}{4}\right)^2 + 1^2} \, dt = \frac{5}{16} \int_0^1 (4 - 3t) e^t \, dt.$$

Now integrating by parts with  $u = 4 - 3t$  and  $dv = e^t dt$  gives

$$\int_C x e^y \, ds = \frac{5}{16} \left[ (4 - 3t) e^t + 3e^t \right]_0^1 = \frac{5(4e - 7)}{16}.$$

6. Consider the change of variables  $u = 3x$ ,  $v = 2y$ . The region  $R = \{(x, y) \mid (3x)^2 + (2y)^2 \leq 1, x, y \geq 0\}$  is the image under this transformation of the region

$$S = \{(u, v) \mid u^2 + v^2 \leq 1, u, v \geq 0\}$$

in the  $uv$ -plane. On the other hand, the Jacobian of this transformation is

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{2} \end{vmatrix} = \frac{1}{6}.$$

These yield  $\iint_R y \, dA = \iint_S \frac{1}{6}(v/2) \, du \, dv$ . Note that  $S$  may be represented in polar coordinates by

$$0 \leq \theta \leq \pi/2, \quad 0 \leq r \leq 1,$$

which give

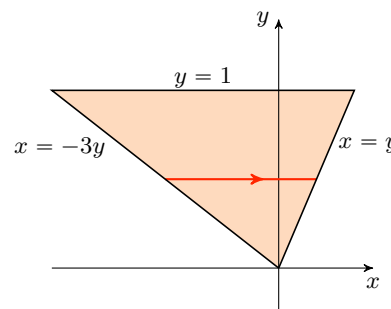
$$\iint_R y \, dA = \frac{1}{12} \int_0^{\pi/2} \int_0^1 r^2 \sin \theta \, dr \, d\theta = \frac{1}{36} \int_0^{\pi/2} \left[ r^3 \right]_0^1 \sin \theta \, d\theta = \frac{1}{36} \int_0^{\pi/2} \sin \theta \, d\theta = -\frac{1}{36} \cos \theta \Big|_0^{\pi/2} = \frac{1}{36}.$$

7. Let  $T$  be the region bounded by the given triangle. Then

$$T = \{(x, y) \mid 0 \leq y \leq 1, -3y \leq x \leq y\},$$

and we have

$$\begin{aligned} A(S) &= \iint_T \sqrt{1 + 2^2 + (-2y)^2} \, dA \\ &= \int_0^1 \int_{-3y}^y \sqrt{5 + 4y^2} \, dx \, dy = \int_0^1 4y \sqrt{5 + 4y^2} \, dx \, dy \\ &= \frac{1}{3} (5 + 4y^2)^{3/2} \Big|_0^1 = \frac{1}{3} (27 - 5\sqrt{5}). \end{aligned}$$



## Math 211 Second Exam (Version B: Sections 2, 3, 51, 52)

7 July 2018

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*Question 1: 10 points, questions 2-7: 15 points each.*

1. Let  $f(x)$  be a positive continuous function on  $[a, b]$  and let  $D$  be the region under the graph  $y = f(x)$  and above the interval  $[a, b]$ . Show that  $\iint_D y \, dA = \int_a^b \frac{1}{2} [f(x)]^2 \, dx$ .
2. Evaluate  $\int_0^4 \int_{y/2}^2 y(1+x^3)^{-\frac{1}{2}} \, dx \, dy$ .
3. Let  $E$  be the solid that lies inside the sphere  $x^2 + y^2 + z^2 = 6z$  and outside the sphere  $x^2 + y^2 + z^2 = 9$ . Give a description of  $E$  in spherical coordinates.
4. Let  $E$  be the solid below the cone  $z = \sqrt{x^2 + y^2}$  and above the ring  $1 \leq x^2 + y^2 \leq 4$  in the  $xy$ -plane. Find the mass of  $E$  if the density function is  $\rho(x, y, z) = x^2$ .
5. Evaluate  $\int_C x e^y \, ds$  where  $C$  is the line segment from  $P(\frac{1}{4}, 0)$  to  $Q(1, 1)$ .
6. Let  $R$  be the region in the first quadrant that is bounded by the ellipse  $\frac{x^2}{9} + \frac{y^2}{4} = 1$ . Evaluate  $\iint_R x \, dA$  by making an appropriate change of variables.
7. Find the area of the part of the surface  $z = 7 + 2x - \frac{1}{2}y^2$  that lies above the triangle with vertices  $(0, 0)$ ,  $(-1, 1)$ , and  $(3, 1)$ .

## SOLUTIONS

1. The region  $D$  can be described as a Type I region:  $D = \{(x, y) \mid a \leq x \leq b, 0 \leq y \leq f(x)\}$ , which gives

$$\iint_D y \, dA = \int_a^b \int_0^{f(x)} y \, dy \, dx = \int_a^b \left[ \frac{1}{2} y^2 \right]_0^{f(x)} dx = \int_a^b \frac{1}{2} [f(x)]^2 \, dx.$$

2. Note that  $\int_0^4 \int_{y/2}^2 y(1+x^3)^{-\frac{1}{2}} \, dx \, dy = \iint_D y(1+x^3)^{-\frac{1}{2}} \, dA$  where

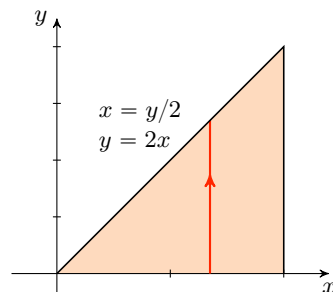
$$D = \{(x, y) \mid 0 \leq y \leq 4, \frac{y}{2} \leq x \leq 2\}.$$

Using a sketch of  $D$  we may describe it as a Type II region:

$$D = \{(x, y) \mid 0 \leq x \leq 2, 0 \leq y \leq 2x\}$$

which gives

$$\iint_D y(1+x^3)^{-\frac{1}{2}} \, dA = \int_0^2 \int_0^{2x} y(1+x^3)^{-\frac{1}{2}} \, dy \, dx = \int_0^2 2x^2(1+x^3)^{-\frac{1}{2}} \, dx = \frac{4}{3} (1+x^3)^{\frac{1}{2}} \Big|_0^2 = \frac{8}{3}.$$



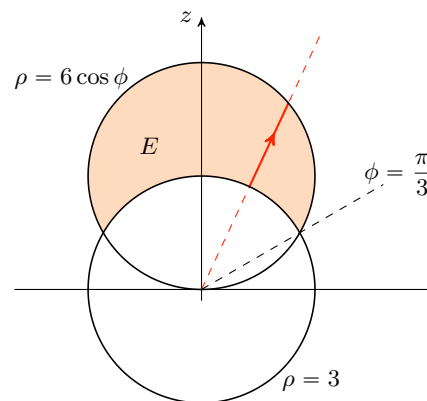
3. A sketch of a vertical cross section of the solid  $E$  is shown.

The two given spheres are represented by the spherical equations  $\rho = 6 \cos \phi$  ( $0 \leq \phi \leq \pi/2$ ) and  $\rho = 3$  respectively, thus their intersection is at

$$6 \cos \phi = 3 \Leftrightarrow \cos \phi = \frac{1}{2} \Rightarrow \phi = \pi/3$$

Now a description of  $E$  in spherical coordinates is

$$E = \{(\rho, \theta, \phi) \mid 0 \leq \theta \leq 2\pi, 0 \leq \phi \leq \frac{\pi}{3}, 3 \leq \rho \leq 6 \cos \phi\}.$$



4. In cylindrical coordinates,  $E = \{(r, \theta, z) \mid 0 \leq \theta \leq 2\pi, 1 \leq r \leq 2, 0 \leq z \leq r\}$ . Thus

$$\begin{aligned} m &= \iiint_E x^2 \, dV = \int_0^{2\pi} \int_1^2 \int_0^r r^3 \cos^2 \theta \, dz \, dr \, d\theta = \int_0^{2\pi} \int_1^2 r^4 \cos^2 \theta \, dr \, d\theta \\ &= \int_0^{2\pi} \left[ \frac{1}{5} r^5 \right]_1^2 \cos^2 \theta \, d\theta = \frac{31}{5} \int_0^{2\pi} \frac{1}{2} (1 + \cos 2\theta) \, d\theta = \frac{31}{5} \left[ \frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right]_0^{2\pi} = \frac{31\pi}{5}. \end{aligned}$$

5. Using the vector  $\overrightarrow{PQ} = \langle \frac{3}{4}, 1 \rangle$ , a parametric representation of the line segment  $C$  is

$$x(t) = (1 + 3t)/4, \quad y(t) = t, \quad 0 \leq t \leq 1,$$

using which we obtain

$$\int_C x e^y \, ds = \frac{1}{4} \int_0^1 (1 + 3t) e^t \sqrt{\left(\frac{3}{4}\right)^2 + 1^2} \, dt = \frac{5}{16} \int_0^1 (1 + 3t) e^t \, dt.$$

Now integrating by parts with  $u = 1 + 3t$  and  $dv = e^t dt$  gives

$$\int_C x e^y \, ds = \frac{5}{16} \left[ (1 + 3t) e^t - 3e^t \right]_0^1 = \frac{5(e + 2)}{16}.$$

6. Consider the change of variables  $u = x/3$ ,  $v = y/2$ . The region  $R = \{(x, y) \mid (x/3)^2 + (y/2)^2 \leq 1, x, y \geq 0\}$  is the image under this transformation of the region

$$S = \{(u, v) \mid u^2 + v^2 \leq 1, u, v \geq 0\}$$

in the  $uv$ -plane. On the other hand, the Jacobian of this transformation is

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 3 & 0 \\ 0 & 2 \end{vmatrix} = 6.$$

These yield  $\iint_R x \, dA = \iint_S 6(3u) \, du \, dv$ . Note that  $S$  may be represented in polar coordinates by

$$0 \leq \theta \leq \pi/2, \quad 0 \leq r \leq 1,$$

which give

$$\iint_R x \, dA = 6 \int_0^{\pi/2} \int_0^1 3r^2 \cos \theta \, dr \, d\theta = 6 \int_0^{\pi/2} \left[ r^3 \right]_0^1 \cos \theta \, d\theta = 6 \int_0^{\pi/2} \cos \theta \, d\theta = 6 \sin \theta \Big|_0^{\pi/2} = 6.$$

7. Let  $T$  be the region bounded by the given triangle. Then

$$T = \{(x, y) \mid 0 \leq y \leq 1, -y \leq x \leq 3y\},$$

and we have

$$\begin{aligned} A(S) &= \iint_T \sqrt{1 + 2^2 + (-y)^2} \, dA \\ &= \int_0^1 \int_{-y}^{3y} \sqrt{5 + y^2} \, dx \, dy = \int_0^1 4y \sqrt{5 + y^2} \, dy \\ &= \frac{4}{3} (5 + y^2)^{3/2} \Big|_0^1 = \frac{4}{3} (6\sqrt{6} - 5\sqrt{5}). \end{aligned}$$

